Toward a four-dimensional Yang–Mills construction

Notebook 31: finite local induction and coupling extraction

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Abstract

We carry the renormalized homogeneous local induction to actual finite periodic regulators. A short-lift Maxwell extraction is normalized by the actual finite classical action. The finite Ward defects are repaired degree by degree; periodic summation by parts and connected localization give a classical contraction with an explicit exponentially small wrapping error. The carrier exponent and field radius are unchanged.

The finite quantum map descends directly to equality classes of normalized total interactions. Its input Lipschitz bound, its quadratic zero-input force, and the finite classical contraction give a uniform block induction. At every intermediate stage an exactly equivalent connected family lies within the next integral's strict budget. The change of representative introduces no approximation error; all raw constants are compensated in the vacuum factor. The number of permitted finite blocks does not enter the amplitude constants.

We also prove evenness in the coupling at fixed retained threshold. A uniform complex coupling disk and Cauchy estimates identify the leading zero-input correction as the pivot term plus the Gaussian determinant, with error bounded by $C[\tau(T) + g^2T^6]$. The physical logarithmic threshold is substituted only after this two-parameter estimate is established.

On the jointly compatible infinite interaction space, a quadratic-only spatial estimate at fixed field radius and the classical semigroup prove exact preservation of the Maxwell coefficient. Thus the coupling drift on an extracted input comes entirely from the quantum remainder. This infinite identity does not set finite cutoff-extraction errors to zero.

These results concern the specified local macrostep and its common analytic continuation. The physical inverse coupling must satisfy the exact extraction recurrence and remain in the weak range. Its drift sign and ultraviolet tuning, the other field sectors, and the continuum observable construction remain to be proved.

The goal remains a nontrivial four-dimensional Yang–Mills quantum theory. This notebook uses the actual finite classical and quantum constructions of Notebooks 15–30, with their fixed gauge, retained window, trace convention, and common domains. It closes the finite representative-budget issue left explicit in Notebook 30.

1 Finite-regulator classical contraction

Fix dimension four, a compact simple gauge group with the fixed color normalization, a primitive odd ratio ℓ, and $0 < \alpha < 1$. A semisimple product uses one reference coefficient per simple factor. Fix the activity exponent $a_0 > 0$ and a grouping J. Let L be an allowed endpoint ratio and write

$$P = LN. \tag{1}$$

Here P is the input current-unit torus period and N is the output current-unit period. Both cubic tori are compatible with the partitions; in particular $J \mid N$. The preceding homogeneous depth M is arbitrary.

On a current torus of period Q, let

$$\|E\|_{Q;a_0,b} = \sup_i \sum_{H \ni i} e^{a_0|H|} \|E_H\|_b, \qquad \Phi_E(A) = \sum_H \big[E_H(A|_H) - E_H(0)\big], \tag{2}$$

where H ranges over nonempty connected sets of J-cells and $\|\cdot\|_b$ is the supremum norm on the complex scalar-coordinate b-polydisc. Quotient by equality of Φ_E on that full polydisc, and denote the quotient norm by $\|[E]\|_Q$. The equality kernel is closed, since finite-volume evaluations are continuous, so this is a Banach quotient. We use its invariant subspace: local gauge, translation, hypercubic and global color identities are imposed on totals. Individual coefficients need not be invariant. Vacuum constants are tracked separately.

1.1 Actual finite inputs and the reference projection

The earlier finite constructions supply the following inputs. First, for $n = 2, 3, 4$, fixed-order Cauchy estimates and connected carriers give, for some fixed $\mu = \mu(J) > 0$,

$$\sup_{b_1} \sum_{b_2,\dots,b_n} e^{\mu D_Q(b_1,\dots,b_n)} |K_{n,Q}(b_1,\dots,b_n)| \le C_n(J) b^{-n} \|E\|_{Q;a_0,b}, \qquad K_{n,Q} = D^n \Phi_E(0). \tag{3}$$

Here D_Q is the maximum pairwise torus ℓ^1 distance between bond midpoints, and internal indices are included in the sums. Indeed, every differentiated tuple lies in a common carrier, $D_Q \le CJ|H|$, while its fixed-degree coordinate count is a polynomial in $D(J)|H|$. Choosing μ small and spending the remaining cardinal reserve absorbs this polynomial without a volume factor.

Second, $S_{M,Q}$ is the actual finite homogeneous classical reference. It has zero value and first derivative at zero and the same total symmetries. Its strict reference family on one fixed radius ρ is uniformly bounded by M_S. Its flat Hessian is the exact relative periodization of the uniformly exponentially decaying q_M, whose midpoint Maxwell coefficient is one.

Removing coefficientwise homogeneous degrees zero and one does not change its normalized total. The phase projection bound gives, for $0 < b \le \rho/2$,

$$\|S^{(\ge 2)}_{M,Q}\|_{Q;a_0,b} \le \frac{M_S (b/\rho)^2}{1 - b/\rho} \le C_S b^2, \qquad C_S = 2M_S/\rho^2. \tag{4}$$

This does not assert that the original individual reference coefficients have zero linear jets.

For $Q \ge 64$, symmetrically lift the quadratic torus kernel using midpoint diameter at most $\lfloor Q/16 \rfloor$, and write its midpoint symbol as $\Gamma^E_{2,Q}$. For distinct directions μ, ν, with the invariant color normalization, put

$$b_Q(E) = \frac{1}{2} \partial^2_{p_\nu} \Gamma^E_{2,Q;\mu\mu}(0). \tag{5}$$

Point symmetry makes the distinct pair immaterial. The exact flat reference periodization implies

$$|b_Q(S_{M,Q}) - 1| \le C(J) e^{-c(J)Q}. \tag{6}$$

This estimate includes both the discarded infinite kernel tail and its long images folded into the short torus lift. Uniformity of the q_M decay makes it uniform in M. Choose $Q_0(J)$ so that the denominator has modulus at least $1/2$, and define

$$\beta_{M,Q}([E]) = \frac{b_Q(E)}{b_Q(S_{M,Q})}, \qquad \mathcal{P}_{M,Q}[E] = [E] - \beta_{M,Q}([E])[S_{M,Q}]. \tag{7}$$

The kernels depend only on the normalized total, so these are well defined on the quotient. Equations (3)–(4) imply

$$|\beta_{M,Q}([E])| \le C_\beta(J) b^{-2} \|[E]\|_Q,$$
$$\beta_{M,Q}([S_{M,Q}]) = 1, \qquad \mathcal{P}_{M,Q}^2 = \mathcal{P}_{M,Q}, \qquad \|\mathcal{P}_{M,Q}\| \le C_{\mathrm{ext}}(J) := 1 + C_\beta(J) C_S. \tag{8}$$

The projection bound, unlike the bound on β alone, is independent of the small b.

Third, the actual finite pre-gauge ordinary representative $a = a_{M,L,N}(w)$ is the partial ordinary average of the full ML-refined ordinary minimizer. The exact finite two-minimum comparison gives

$$a(0) = 0, \qquad S_{M,LN}(a(w)) = S_{ML,N}(w). \tag{9}$$

The identity first holds for real small fields and then holomorphically. In dimension four it has no additional dimensional prefactor.

For any fixed finite cardinal reserve A, this map has coherent complete-field support coefficients on final unit cells:

$$a|_i = \sum_{\varnothing \ne X} a_{i,X}(w_X), \qquad a_{i,X}(0) = 0,$$
$$\sup_i \sum_X e^{\kappa \tau_N(\{i\} \cup X) + A|X|} \|a_{i,X}\|_{C^{1,\alpha};b} \le C_* b/L. \tag{10}$$

The whole current-field output fiber and its physical gradient Hölder quotients are inside the support sum. Coefficients on overlapping fixed collars are restrictions of one global assignment. The constants and the permitted classical radius are uniform in M, L, N. This is the finite complete-field prefix theorem, not a periodization assumption about an infinite background.

Define scalar classical substitution by

$$(\mathcal{T}_{M,L,N} \Phi_E)(w) = \Phi_E(a_{M,L,N}(w)). \tag{11}$$

For L sufficiently large, the amplitude is strictly inside the input b-domain, so equality of finite totals implies equality after substitution. The estimates below give a bounded output representative.

We also record the finite-domain invariance argument. For a real output field and a sufficiently small local gauge transform, the finite transfer and gauge-flow construction gives an actual near-identity connecting input gauge g. Both endpoint angles have size $\rho_0 = C_A b/L$. The bond equation $V^g = U$ bounds the edge increments of g by $C(|U - I| + |V - I|)$, and hence $|D \log g| \le C\rho_0$ on its fixed near-identity chart. The existence of the small chart comes from that actual finite construction, not merely from an edge bound. Every field along $\exp(t \log g)$, $0 \le t \le 1$, has scalar-coordinate size at most $C_g \rho_0$. Fixed matrix/color norm conversions are included in C_g. Taking $L > C_g C_A$ with a strict margin keeps the entire path inside b. The input Ward identity integrates along it, and analytic continuation gives the local complex output Ward identity. No infinite gauge truncation is needed. The finite minimum comparisons or exact covariance likewise give the spatial and global color identities on this common domain.

Consequently $\mathcal{T}$ acts on the finite invariant quotients. The full-reference identity gives exactly

$$\mathcal{T}_{M,L,N}[S_{M,LN}] = [S_{ML,N}], \qquad \mathcal{P}_{ML,N} \mathcal{T}_{M,L,N} \mathcal{P}_{M,LN} = \mathcal{P}_{ML,N} \mathcal{T}_{M,L,N}. \tag{12}$$

This does not assert that $\mathcal{T}$ preserves the zero-β subspace.

1.2 Graded Ward defects and finite-range jet repair

Let $[F]$ satisfy $\beta_{M,P}([F]) = 0$, and choose a representative of norm at most ε, as close to its quotient norm as desired. Set $T_0 = \lfloor P/16 \rfloor$. The symmetric infinite lift

$$\widetilde{K}_n(b_1, \dots, b_n) = K_{n,P}(b_1 \bmod P, \dots, b_n \bmod P)\mathbf{1}_{\{D_\infty(b_1,\dots,b_n) \le T_0\}} \tag{13}$$

preserves slot symmetry, midpoint parity and the spatial and color symmetries. A short torus tuple has a unique lift with its first bond fixed: all coordinate differences are below half a period, and the lifted pairwise distances are the torus distances. Thus

$$K_{n,P} = \mathrm{Per}_P \widetilde{K}_n + K_n^{\mathrm{tail}}, \qquad K_n^{\mathrm{tail}} = K_{n,P}\mathbf{1}_{\{D_P > T_0\}}. \tag{14}$$

Every fixed spatial moment of the tail is bounded by $C(J,m)\varepsilon b^{-n} e^{-c(J)P}$.

Ward operations are finite sums of nearest-neighbor shifts, coincident-slot insertions and color commutators. They commute with relative periodization. A lifted Ward defect has diameter at most $T_0 + 2$, where periodization is injective. Its infinite fixed moments are therefore bounded by those of the discarded tails.

Keep the field-degree grading. For $b \le 1$, the degree-n identity involves only degrees $2, \dots, n$, so its fixed-moment error is bounded by

$$\delta_n = C(J)\varepsilon b^{-n} e^{-c(J)P}, \qquad n = 2, 3, 4. \tag{15}$$

The quadratic coefficient elimination uses only δ_2; the cubic uses δ_3 and the already bounded lower jets, and the quartic uses δ_4 and its lower jets. Since $b \le 1$, these lower terms fit the displayed grading. A common b^{-4} estimate for all degrees is neither needed nor used.

For completeness, the lifted hypercubic Hessian has mass m and quadratic coefficients a, b, c. The degree-one and degree-three Ward comparisons bound m, a, c by $C\delta_2$, because the extracted coefficient $b_P(F)$ is exactly zero. Odd midpoint orders vanish by parity. Write the cubic linear polynomial as $Ap + Bq$; the independent $p_\mu q_\sigma$ Ward coefficients bound B, and the $q_\nu p_\sigma$ coefficients bound A, by $C\delta_3$. The quartic leading identity contracts its constant tensor with an arbitrary gauge momentum and bounds it by $C\delta_4$. Hence the lifted quadratic jets through degree three, cubic jets through degree one, and quartic constant are respectively $O(\delta_2), O(\delta_3), O(\delta_4)$.

Pass to tail-coordinate Laurent symbols $F_n(z)$, anchoring the first positive-bond tail and retaining internal indices. Fixed midpoint phases and the local change $z = e^{ip}$ preserve these finite-order bounds. Put $m_2 = 3, m_3 = 1, m_4 = 0$, and take

$$J_{m_n} F_n(z) = \sum_{|\gamma| \le m_n} \frac{\partial_z^\gamma F_n(1)}{\gamma!}(z-1)^\gamma. \tag{16}$$

Its kernel is of fixed finite relative range and has absolute row norm at most $C\delta_n$.

Average this kernel over the $n!$ slot permutations, reanchoring the first slot each time, and call the result R_n. Its relative range increases only by a fixed factor and its norm remains $O(\delta_n)$. The corrected kernel

$$\widehat{K}_n = \widetilde{K}_n - R_n \tag{17}$$

has exactly zero jets through degree m_n. Indeed each reanchored residual is the composition of $F_n - J_{m_n}F_n$ with an analytic monomial change fixing $z = 1$, possibly times a harmless analytic phase. This preserves vanishing order $m_n + 1$; averaging preserves it as well. Reality and color covariance can be retained. Point symmetry of the repair is not required by the spatial estimate.

For a fixed smaller $\sigma = \sigma(J) > 0$,

$$\sup_{\text{first index}} \sum_{\text{others}, r_2,\dots,r_n} e^{\sigma \sum_{j\geq 2} |r_j|} |\widehat{K}_n(0, r_2, \dots, r_n)| \leq C(J)\varepsilon b^{-n},$$

$$\|R_n\|_{\text{row}} \leq C(J)\varepsilon b^{-n} e^{-c(J)P}. \tag{18}$$

The second line has fixed finite relative range. The finite polynomial decomposition is exact:

$$\frac{K_{n,P}(a,\dots,a)}{n!} = \frac{(\mathrm{Per}_P\, \widehat{K}_n)(a,\dots,a)}{n!} + \frac{(\mathrm{Per}_P\, R_n)(a,\dots,a)}{n!} + \frac{K_n^{\text{tail}}(a,\dots,a)}{n!}. \tag{19}$$

No error term is discarded or separately asserted to be gauge invariant.

1.3 Direct periodic spatial localization

We give the finite version of the spatial argument explicitly. For a relative infinite kernel $\widehat{K}_n$, its periodized polynomial equals

$$\frac{1}{n!} \sum_{x\in T_P} \sum_{r_2,\dots,r_n \in \mathbb{Z}^4} \widehat{K}_n(r_2,\dots,r_n)[a(x), a(x+r_2), \dots, a(x+r_n)], \tag{20}$$

where fields are read periodically. The relative sums converge absolutely, and repeated torus slots retain their multilinear multiplicity.

For the quadratic symbol $F(z) = \sum_r \widehat{K}_2(r) z^r$, define

$$B_j F(z) = \frac{F(1,\dots,1,z_j,\dots,z_d) - F(1,\dots,1,1,z_{j+1},\dots,z_d)}{z_j - 1}.$$

The quotient is removable, and $F - F(1) = \sum_j (z_j - 1) B_j F$. A monomial contributes $|r_j|$ terms of exponent length at most $|r|_1$. Thus the combined row Wiener norm, including the j sum, loses at most $[e(\sigma - \sigma')]^{-1}$. Vanishing order decreases by at most one. Apply twice and put $\widetilde{H}_{\rho\tau} = -z_\rho B_\tau B_\rho F$. Then

$$F = \sum_{\rho,\tau} (z_\rho^{-1} - 1)\widetilde{H}_{\rho\tau}(z)(z_\tau - 1), \qquad \widetilde{H}_{\rho\tau}(1) = 0. \tag{21}$$

The combined row norm at exponent $\sigma/2$ is at most

$$\frac{16 e^{\sigma/2}}{e^2 \sigma^2} C(J)\varepsilon b^{-2}.$$

Finite periodic summation by parts has no boundary term. Equation (21) and the zero sum of $\widetilde{H}$ give exactly

$$\frac{1}{2} \sum_{\substack{x\in T_P, r \\ \rho,\tau}} [D_\rho a(x)]^T \widetilde{H}_{\rho\tau}(r)[D_\tau a(x+r) - D_\tau a(x)]. \tag{22}$$

No summability of an infinite field is assumed.

For the cubic term set $\ell_r(x) = \sum_\rho r_\rho D_\rho a(x)$ and $\mathrm{Rem}_r = a(x+r) - a(x) - \ell_r(x)$. The exact lifted constant and first moments cancel the all-anchor and single-ℓ terms. The six remaining terms are the two a, a, Rem terms, a, ℓ, ℓ, the two a, ℓ, Rem terms, and $a, \mathrm{Rem}, \mathrm{Rem}$. For the quartic term subtract the all-anchor product and telescope its three shifted factors. These are identities for periodic fields with lifted integer offsets; no periodic affine function is introduced.

Let $h = L^{-1}$ and $m = \sum_{j\ge 2} |r_j|_1$. For $m \le L$, coherence in (10) gives, in the weighted family norm on one fixed enlarged collar,

$$\begin{aligned} |a| &\le Cbh, & |Da| &\le Cbh^2, \\ |Da(x+r) - Da(x)| &\le Cbh^{2+\alpha}|r|^\alpha, & |\operatorname{Rem}_r| &\le Cbh^{2+\alpha}|r|^{1+\alpha}. \end{aligned} \tag{23}$$

Negative path traversals use the positive difference at the lower endpoint. Products of coefficient families glue their support trees and preserve cardinal reserve. The quadratic, cubic and quartic residual products are therefore bounded respectively by

$$Cb^2h^{4+\alpha}|r|^\alpha, \qquad Cb^3h^{4+\alpha}(1+m)^6, \qquad Cb^4h^5(1+m).$$

The kernel moments cost only fixed powers of σ^{-1}, and their b^{-n} cancels b^n.

For every product, project the finite endpoint root set and every terminal set $\{\text{root}\} \cup X$ to final J-cells. Choose the short connecting graphs supplied by the projection lemma and take their connected union H. The carrier need not contain the full projection of an arbitrarily chosen fine tree. With a fixed number n_0 of endpoint roots,

$$\begin{aligned} &|H| \le 1 + T_H, \\ &T_H \le \frac{2}{J}\left[\tau_N(\text{roots}) + \sum_{\text{occ}} \tau_N(\{\text{root}\} \cup X)\right] + 4\left[n_0 + \sum_{\text{occ}} |X|\right]. \end{aligned} \tag{24}$$

Here τ_N uses final unit cells of period N. Shortest periodic paths obey the same bound; wrapping can only reduce the union cardinality. If $j(x)$ is the final J-cell containing x/L, then

$$e^{a_0|H|}\mathbf{1}_{\{z\in H\}} \le e^{a_0} e^{-\operatorname{dist}_N(z,j(x))} e^{(a_0+1)T_H}. \tag{25}$$

The distance here is on final J-cells, of period N/J, not on unit cells. Choose

$$A \ge 4(a_0+1), \qquad 2(a_0+1)/J \le \kappa. \tag{26}$$

These pay the support and attached-tree weights. Finally,

$$\sum_{x\in T_P} e^{-\operatorname{dist}_N(z,j(x))} \le (JL)^4 S_1, \qquad S_1 = \sum_{y\in\mathbb{Z}^4} e^{-|y|_1}. \tag{27}$$

There are exactly $(JL)^4$ current anchor sites per output J-cell. Choosing a shortest representative per torus coset bounds the remaining sum by S_1.

For $m > L$, use separate endpoint roots. Their joining tree has length at most $C(m/L+1)$, so (25) adds at most $Ce^{C'm/(JL)}$. The rough quadratic, cubic and quartic residual bounds are respectively Cb^2h^4, $Cb^3h^3(1+m)^2$, and Cb^4h^4. Taking L large after J makes $C'/(JL) \le \sigma/8$. The unspent kernel decay bounds all far terms by

$$C(J)\varepsilon(L+2)e^{-\sigma L/8} \le C'(J)\varepsilon L^{-\alpha}.$$

The near terms have the same or better bound after (27). Normal regrouping by H therefore constructs a connected output family with

$$\left\| \sum_{n=2}^{4} \frac{(\operatorname{Per}_P \widehat{K}_n)(a,\dots,a)}{n!} \right\|_{\text{rep}} \le C_{\text{low}}(J)\varepsilon L^{-\alpha}. \tag{28}$$

The notation $\|\cdot\|_{\text{rep}}$ denotes the norm of the constructed representative, hence bounds its quotient norm. Its coefficients are holomorphic on the same b-polydisc. This proof used the actual finite periodic support family, not a putative infinite-volume extension of it.

1.4 Wrapping errors at the unchanged activity exponent

We use a direct endpoint-product bound. Suppose a torus n-kernel W, $n \le 4$, satisfies

$$\sup_{b_1} \sum_{b_2,\dots,b_n} e^{\mu D_P} |W| \le A_n. \tag{29}$$

Join each endpoint to the first by a shortest torus path. Its root tree on final unit cells has length at most $C_n(D_P/L+1)$. Insert the actual endpoint support families and use (24)–(25). The additional factor is at most

$$C \exp\!\left(\frac{C_n'(a_0+1)D_P}{JL}\right).$$

Increase the lower threshold for L until this exponent coefficient is at most $\mu/4$. The field product contributes $(Cb/L)^n$, while (27) gives

$$\begin{aligned} \left\| \frac{W(a,\dots,a)}{n!} \right\|_{\mathrm{rep}} &\le C(J,n) A_n b^n L^{4-n}, \\ \left\| \frac{(W\mathbf{1}_{\{D_P>T_0\}})(a,\dots,a)}{n!} \right\|_{\mathrm{rep}} &\le C(J,n) A_n b^n L^{4-n} e^{-\mu T_0/2}. \end{aligned} \tag{30}$$

The target exponent is exactly a_0. The extra carrier length was paid after spatial compression by JL, rather than by lowering the old-scale activity exponent.

Apply the second inequality to K_n^{tail}, using $A_n \le C(J)\varepsilon b^{-n}$ and $T_0 \ge P/32$. Apply the first to the fixed-range repair R_n, retaining its factor $e^{-c(J)P}$ from (18). Relative periodization counts every occurrence, including any repeated periodic slots. Together these give

$$\left\| \sum_{n=2}^{4} \frac{(\mathrm{Per}_P\, R_n + K_n^{\mathrm{tail}})(a,\dots,a)}{n!} \right\|_{\mathrm{rep}} \le C_{\mathrm{err}}(J)\varepsilon(L^2+L+1)e^{-c(J)LN}. \tag{31}$$

All b-powers cancel in their correct degree. An old-scale tail estimate that loses part of a_0 is not being invoked.

1.5 High degrees and the finite quotient estimate

Coefficientwise homogeneous phase projection has norm at most one. Thus

$$R_{\ge 5}E = E - \sum_{n=0}^{4} P_n E, \qquad \|R_{\ge 5}E\| \le 6\|E\|. \tag{32}$$

The supported fifth-order background theorem gives

$$\|\mathcal{T}_{\mathrm{high}} R_{\ge 5}E\|_{N;a_0,b} \le 6C_{\mathrm{high}}\varepsilon/L, \qquad C_{\mathrm{high}} = C_0(8C_A)^5. \tag{33}$$

After the fixed classical reserves, C_{high} does not depend on M, L, J, N. A fixed J satisfying the earlier tail budgets can be used while increasing the endpoint to $L = R^q$: the core-compression coefficient decreases, whereas the $D(J)t$ and projection budgets are unchanged. The field-scaling disk has radius $L/(8C_A)$; choose b afterward so that

$$b \le 1, \quad b \le \rho/2, \quad bL/(8C_A) \le r_A, \quad L/(8C_A) \ge 2, \tag{34}$$

and impose the other fixed classical domain restrictions.

The degree-zero sector disappears from the normalized total. The degree-one total vanishes by semisimple global color invariance and remains zero after substitution, even if its individual coefficients are nonzero. The degree-two through degree-four polynomials are exactly (19). Thus the low estimate, error estimate, and high estimate represent the entire finite normalized substituted total, not merely its first few kernels.

Theorem 1.1 (Finite classical contraction)**.** *Fix J and its stated classical tail and support budgets. There are $L_0(J)$, $N_0(J)$, positive $c(J)$, and constants $C_{\mathrm{low}}(J)$, $C_{\mathrm{err}}(J)$, $C_{\mathrm{ext}}(J)$, independent of L, b, M, N, such that every allowed $L \geq L_0(J)$, every compatible $N \geq N_0(J)$, and every sufficiently small b chosen after L satisfy*

$$\begin{aligned}&\|\mathcal{P}_{ML,N}\mathcal{T}_{M,L,N}\mathcal{P}_{M,LN}\| \\ &\quad \leq C_{\mathrm{ext}}(J)^2\left[C_{\mathrm{low}}(J)L^{-\alpha} + \frac{6C_{\mathrm{high}}}{L} + C_{\mathrm{err}}(J)(L^2+L+1)e^{-c(J)LN}\right].\end{aligned} \tag{35}$$

The input and output finite invariant quotients have the same exponent a_0 and radius b. Separate input and output projection constants can replace the square if their conventions differ.

Proof. Take an almost-minimal representative of the extracted class. Its norm can approach $\|\mathcal{P}_{M,LN}[E]\| \leq C_{\mathrm{ext}}(J)\|[E]\|$. Apply (28), (31), and (33) to this representative. They give an exactly equivalent connected output family. The output projection costs the second $C_{\mathrm{ext}}(J)$. Take the quotient infimum and send the representative error to zero. □

Renaming constants, the bound has the form

$$\|\mathcal{P}_{\mathrm{out}}\mathcal{T}\mathcal{P}_{\mathrm{in}}\| \leq A(J)L^{-\alpha} + B(J)/L + E(L,J)e^{-c(J)N}. \tag{36}$$

The sharper (35) displays the actual input period LN. For any desired $\theta > 0$, choose a finite allowed power $L = R^q$ after the fixed J to make the first two terms at most $\theta/2$; enlarge the compatible threshold N_0 until the error is at most $\theta/2$ for all $N \geq N_0$; and choose b afterward. This is a θ-contraction uniform in preceding depth and all these finite regulators. Shrinking b does not change its constants.

By (12) the same bound applies to $\mathcal{P}_{\mathrm{out}}\mathcal{T}$, with the input extraction implicit. That identity uses the full actual finite reference, not merely an agreement of flat Hessians.

1.6 Scope

The estimate uses actual finite normalized totals and periodic spatial algebra. It requires neither an infinite-volume interaction limit nor evaluation of an infinite total at a nonsummable periodic field. Small finite lifted Ward defects and wrapping tails are retained explicitly. The resulting small representatives are exactly equivalent on the full finite b-polydisc; the original carrier assignment need not itself obey the improved norm.

The order is fixed J, then a finite endpoint $L = R^q$, then the small b. This removes the finite-regulator classical relocalization issue. It does not identify q sequential quantum cutoff integrals with one direct macro cutoff integral. Application to an actual finite quantum recursion still needs the exact finite scalar descent, every intermediate analytic and Gaussian budget, the running-coupling bounds and the specified cutoff/history identities. Raw vacuum normalizations remain separate. No beta-function sign, indefinitely iterable estimate, large-field control or continuum existence claim follows from this classical result alone.

2 Exact finite local quantum induction

This section combines the actual common-domain finite macro integral, finite-total quotient descent, the fixed-outer-radius force of Notebook30, and Theorem 1.1. The result concerns the selected homogeneous *local* macrostep and its specified holomorphic continuation. It does not identify the full Wilson pushforward, other field sectors, or arbitrary mixed histories.

2.1 Finite geometry, totals and the actual map

Fix the primitive ratio ℓ, an elementary composite ratio $R = \ell^s$, auxiliary grouping J, and exponent $a_0 = 1/16$ satisfying the strict common-domain budgets. At stage j let

$$M_j = M_0 R^j, \qquad Q_j = Q_0/R^j. \tag{37}$$

The preceding homogeneous classical depth is M_j; the cubic current-unit torus period is Q_j, after resetting spacing to one. All periods are compatible integers and $J \mid Q_j$. Constants may depend on the fixed geometry and color conventions, but are uniform in depth and in the permitted periods above a lower threshold.

On the period-Q torus use the finite activity space and quotient of (2). For clarity, the raw and normalized totals are

$$E^{\mathrm{raw}}(A) = \sum_H E_H(A|_H), \qquad \Phi_E(A) = E^{\mathrm{raw}}(A) - E^{\mathrm{raw}}(0), \qquad \|E\|_{Q;a_0,b} = \sup_i \sum_{H \ni i} e^{a_0|H|} \|E_H\|_b. \tag{38}$$

There are finitely many coordinates and connected J-cell carriers at each regulator. Equality of Φ_E on the full product ball is a closed linear kernel, so the quotient is Banach. We impose local gauge, translation, hypercubic and global color identities on totals, not individual coefficients. Use a simple Lie algebra for the scalar Maxwell functional; a finite componentwise version handles a semisimple product.

Let $S_{M,Q}$ be the actual finite homogeneous ordinary minimum action, with $S_{M,Q}(0) = 0$. It is not replaced by a canonical periodization of an independently selected infinite reference. The actual common-domain construction supplies

$$\begin{aligned} \mathcal{F}_j(E) &= \mathcal{T}_j E + \mathcal{G}_j(E), \\ \mathcal{G}_j(E) &= \Delta_j + \mathcal{Q}_{\mathrm{det},j} + \mathcal{L}_j(Q_E). \end{aligned} \tag{39}$$

Here $\mathcal{T}_j$ is pre-gauge substitution, Q_E is the exact inherited fluctuation difference, and $\mathcal{L}_j$ is the strict Gaussian logarithm with the fixed actual base exponent, precision, retained window and chart. The coupling g_j and threshold $T_{\mathrm{cut},j}$ are held fixed when comparing inputs to this one map. Field-independent factors are tracked separately.

On the original real common overlap, this is the exact chosen local macro integral. On the common radius-b ball it is its proved holomorphic continuation, evaluated through pre-gauge fields inside the old activity domain. Identification throughout that ball with a globally defined gauge-orbit density requires the extra orbit-representation hypothesis of the common-domain theorem. It is not inferred from a quotient identity.

2.2 Exact finite quotient descent

Lemma 2.1 (Finite descent of the actual quantum update)**.** *The actual finite maps $\mathcal{F}_j$, $\mathcal{T}_j$ and $\mathcal{G}_j$ descend to the invariant finite quotient on their admissible ball. No infinite periodization, kernel limit or translation averaging is required.*

Proof. If $[E] = [E']$, then on the entire input product ball

$$E^{\mathrm{raw}}(A) - E'^{\mathrm{raw}}(A) = c, \qquad c = E^{\mathrm{raw}}(0) - E'^{\mathrm{raw}}(0). \tag{40}$$

The actual pre-gauge field $a_p(w)$ and all fluctuation fields $f_\lambda(w, \xi)$ used in the inherited construction lie in that ball on the fixed retained window and the established interpolation domains. Consequently

$$E^{\mathrm{raw}}(f_1) - E^{\mathrm{raw}}(f_0) = E'^{\mathrm{raw}}(f_1) - E'^{\mathrm{raw}}(f_0). \tag{41}$$

The scalar inherited Gaussian insertion is identical, while the exterior background terms differ by c. The covariance, base action, chart, sharp window and measure factors are unchanged. The exact finite integrals therefore differ only by e^c. Their normalized holomorphic logarithms agree on the connected domain, using the common nonvanishing branch. A raw logarithm constant belongs to the vacuum normalization and disappears from the quotient. The localized arrays need not agree.

Classical substitution descends immediately from (40) evaluated at $a_p(w)$; subtracting it gives descent of $\mathcal{G}_j$. The exact real-sector symmetry and holomorphic Ward continuation preserve the stipulated total identities. No carrierwise gauge identity is assumed. □

The constant in (40) must be compensated in a raw density, as specified below. For a real class, a chosen representative can be made real without increasing its norm by averaging $E_H(w)$ with $\overline{E_H(\overline{w})}$.

2.3 Uniform finite difference and forcing estimates

Let $\bar{\epsilon}$ be the family ceiling of the strict two-input comparison. For quotient inputs x, y of norm at most $\bar{\epsilon}/4$, choose near-minimal representatives Y of y and D of $x - y$, and put $X = Y + D$. For arbitrarily small $\delta > 0$,

$$\begin{aligned} \|Y\| &\le \bar{\epsilon}/4 + \delta, \\ \|D\| &\le \|x - y\| + \delta \le \bar{\epsilon}/2 + \delta, \\ \|X\| &\le 3\bar{\epsilon}/4 + 2\delta < \bar{\epsilon}. \end{aligned} \tag{42}$$

Both complete inputs are admissible and have invariant totals. Covariance of their arrays is unnecessary at finite volume.

The actual Gaussian interpolation retains the base exponent and precision. In particular impose

$$v \le \frac{\eta_*}{64e}, \qquad a_Q \bar{\epsilon} \le \frac{\eta_*}{128e}, \qquad a_Q = 16 C_T C_R \frac{g_j T_{\mathrm{cut},j}}{b}. \tag{43}$$

The inherited difference is linear in D. The finite Banach-valued logarithm Cauchy estimate gives

$$\|\mathcal{G}_j(x) - \mathcal{G}_j(y)\|_{Q_{j+1}} \le L_{\mathrm{quant},j} \|x - y\|_{Q_j}, \qquad L_{\mathrm{quant},j} = C_{\mathrm{quant}} \frac{g_j T_{\mathrm{cut},j}}{b}, \quad C_{\mathrm{quant}} = 1024 e S_1 C_T C_R. \tag{44}$$

Apply the family bound to the compatible X, Y, use Lemma 2.1, and let δ decrease to zero. No Fatou or local-limit argument is needed at a fixed regulator. The external determinants cancel in this difference, and no equal-low-jet hypothesis is used.

For the zero input, choose one outer field radius ρ after R, J with all strict classical and Gaussian margins. Fix the same actual threshold, charts, covariance and branch before restricting to b. Notebook30's finite outer-domain force has a uniform anchored bound H_* on ρ. Its total linear jet vanishes by semisimple adjoint invariance. Removing coefficientwise degrees zero and one and using homogeneous phase projection gives, for $b \le \rho/2$,

$$\|\mathcal{G}_j(0)\|_{Q_{j+1}} \le C_F b^2, \qquad C_F = 2H_*/\rho^2. \tag{45}$$

This is an exactly equivalent normalized representative, not a low-jet approximation. Its vacuum constant is kept separately. The full force need not be $O(g^2)$: the pivot and Gaussian determinant terms can persist at zero coupling.

2.4 Finite references and the classical block

At stage j, use the finite Maxwell ratio of (7), normalized against the actual reference $S_j = [S_{M_j,Q_j}]$. For periods above its fixed threshold,

$$\beta_j(S_j) = 1, \qquad \|\beta_j\| \le C_\beta b^{-2}, \qquad P_j = I - S_j\beta_j, \qquad \|P_j\| \le p := 1 + C_\beta C_S, \qquad \|S_j\| \le C_S b^2. \tag{46}$$

The b^{-2} factor is retained and cancels only in the combined projection estimate. All constants are uniform in the permitted periods and preceding depth.

The actual finite two-minimum comparison and ordinary composition give

$$\mathcal{T}_j S_j = S_{j+1}, \qquad P_{j+1}\mathcal{T}_j P_j = P_{j+1}\mathcal{T}_j. \tag{47}$$

In dimension four the fine action has the identical $\eta^{d-4} = 1$ normalization in both unit conventions. The identity concerns the full actual action, not merely its quadratic symbol. Put $A_j = P_{j+1}\mathcal{T}_j$. The finite classical semigroup for invariant totals yields

$$A_{j+q-1}\cdots A_j P_j = P_{j+q}\mathcal{T}_{M_j,R^q,Q_{j+q}} P_j. \tag{48}$$

The direct map on the right is classical substitution. It does not replace the intervening quantum integrals or their sharp windows by a one-shot integral.

Write $L = R^q$. Theorem 1.1 gives the block bound

$$\theta(q,N) := p^2\left[C_{\mathrm{low}}(J)L^{-\alpha} + \frac{6C_{\mathrm{high}}}{L} + C_{\mathrm{err}}(J)(L^2+L+1)e^{-c(J)LN}\right], \qquad N = Q_{j+q}. \tag{49}$$

It retains the graded finite Ward defects and wrapping tails at the same a_0, b. Fix J, choose a finite q, and then a compatible N_0 above all reference, collar and period thresholds so that

$$\theta(q,N) \le 1/4 \qquad (N \ge N_0). \tag{50}$$

Only afterward choose b within the direct L requirements of (34) and all common domains. One-step transport has uniform bounds

$$\|\mathcal{T}_j\| \le B_T, \qquad \|A_j\| \le B, \qquad B \ge \max(1, pB_T). \tag{51}$$

The finite projection telescope remains exact although the period and functional β_j change at each stage. It requires no identity $\beta_{j+1}\mathcal{T}_j = \beta_j$.

2.5 A uniform finite block envelope

Define the extracted maps

$$\Psi_j(x) = P_{j+1}\mathcal{F}_j(x) = A_j x + H_j(x), \qquad H_j = P_{j+1}\mathcal{G}_j. \tag{52}$$

Let ν bound $pL_{\mathrm{quant},j}$ over the step data. Equations (44)–(45) give

$$\operatorname{Lip} H_j \le \nu, \qquad \|H_j(0)\| \le C_0 b^2, \qquad C_0 = pC_F. \tag{53}$$

Set

$$K = B + 1, \qquad L_q = \sum_{l=0}^{q-1} K^l, \qquad K_{\mathrm{amp}} = \max\{1, 2C_0 L_q\}, \qquad C_{\mathrm{int}} = K^q K_{\mathrm{amp}} + C_0 L_q. \tag{54}$$

Choose the common b so that $C_{\mathrm{int}} b^2 < \bar{\epsilon}/4$, and finally restrict the weak-coupling range so every strict budget holds and

$$\nu \le \min\{1, (4qK^{q-1})^{-1}\}. \tag{55}$$

For example, a sufficiently large fixed multiple of $\sqrt{\log(1/g)}$ with a fixed lower threshold has $gT(g)/b \to 0$, $gT(g)^3 \to 0$ and $\tau(T(g)) \to 0$. The outer force remains uniform on this smaller coupling range.

Theorem 2.2 (Finite local induction with exact small representatives)**.** *Use the actual finite maps, references and strict domains above. Choose q, N_0, b and the weak-coupling range in the stated order. For every compatible finite history of n steps with*

$$Q_0 = R^n Q_n, \qquad Q_n \geq N_0, \tag{56}$$

and initial extracted class satisfying

$$\beta_0(x_0) = 0, \qquad \|x_0\| \leq r_b := K_{\mathrm{amp}} b^2, \tag{57}$$

the following conclusions hold for couplings in the permitted weak range.

Every full q-step block preserves the radius-r_b extracted ball and has Lipschitz constant at most $1/2$ when compared with the same prescribed coupling data. Every intermediate class has norm at most $C_{\mathrm{int}} b^2 < \bar{\epsilon}/4$. At every stage one can choose an exactly equivalent connected finite representative on the same a_0, b with norm at most $2C_{\mathrm{int}} b^2 < \bar{\epsilon}/2$. Thus every actual next input and Gaussian budget remains valid. The changes of raw constant are compensated by the exact vacuum factors below.

The constants are independent of the permitted number of blocks and of preceding depth. The result describes the selected homogeneous local germ and its continuation; its global measure interpretation retains the stated extra orbit-domain requirements.

Proof. For a block starting at stage j with an input in (57), the one-step bound gives, inductively for $0 \leq k \leq q$,

$$\|x_{j+k}\| \leq K^k K_{\mathrm{amp}} b^2 + C_0 b^2 \sum_{l<k} K^l \leq C_{\mathrm{int}} b^2 < \bar{\epsilon}/4. \tag{58}$$

This verifies the next map's domain before that map is invoked. It also applies to the zero-started comparison trajectory.

For two trajectories using the same couplings, variation of constants bounds their difference by

$$\begin{aligned} \|\delta_q\| &\leq \left[\theta(q, Q_{j+q}) + \nu \sum_{k=0}^{q-1} B^{q-1-k} (B+\nu)^k\right] \|\delta_0\| \\ &= \left[\theta(q, Q_{j+q}) + (B+\nu)^q - B^q\right] \|\delta_0\| \\ &\leq \left[\frac{1}{4} + q\nu K^{q-1}\right] \|\delta_0\| \leq \frac{1}{2} \|\delta_0\|. \end{aligned} \tag{59}$$

Only the initial full classical product uses (49); every shorter product uses powers of B. The block output at zero has norm at most $C_0 L_q b^2 \leq r_b/2$. Comparing with that output therefore proves invariance of the radius-r_b ball. The output is extracted at every step by definition.

Repeat by full blocks; any final partial block satisfies (58). Condition (56) ensures every full block output has the lower period required by (50). On each fixed starting torus there are finitely many permitted steps. Uniformity as that torus grows does not assert an infinite sequence on one fixed finite regulator.

For representative selection, choose at stage j a finite family E_j within $C_{\mathrm{int}} b^2$ of the quotient infimum. Then

$$\|E_j\|_{Q_j; a_0, b} \leq 2C_{\mathrm{int}} b^2 < \bar{\epsilon}/2. \tag{60}$$

The infimum need not be attained. This family is holomorphic on the full same b ball, has connected carriers at the same exponent, and has exactly the required normalized total. Its total symmetries are unchanged. A real class permits the norm-nonincreasing real averaging described earlier.

Use E_j as the next input. Its norm is strictly inside the actual family ceiling, so the inherited support estimates and Gaussian theorem apply. Lemma 2.1 shows that this choice changes no next

quotient output. After the next step and actual finite reference subtraction, choose another such family. Every generated correction remains in the total. Since equality is exact, the positive near-minimal tolerance creates no accumulating approximation error and need not form a summable sequence. The raw constant is handled by (64) below. □

This selection occurs at each finite regulator; it is not an inference from a small infinite class. Every winding contribution is part of the finite total estimated by Theorem 1.1 and the finite quantum bounds. The assertion does not say that an untouched canonical cluster array remains small. Selection is existential in a finite Banach quotient, not a proposed algorithm for finding a minimizing family. Exact equality and the uniform norm are the properties needed for this induction.

The envelope also applies along an adaptively updated physical coupling sequence for as long as it stays weak. For the bound on a single such history one may freeze its realized coupling values and use the auxiliary zero-started comparison with those same values. This does not give a difference contraction between histories that have different updated couplings.

2.6 Exact vacuum compensation and coupling bookkeeping

Write the finite input density on the common chart as

$$Z_j \exp[-\gamma_j S_j + E_j^{\mathrm{raw}}], \qquad \gamma_j = g_j^{-2}. \tag{61}$$

The finite exact chosen local step, with its known scalar normalization N_j, produces

$$Z_j N_j \exp[-\gamma_j S_{j+1} + F_j^{\mathrm{raw}}]. \tag{62}$$

Let

$$\beta_{\mathrm{out}} = \beta_{j+1}([F_j^{\mathrm{raw}}]), \qquad x_{j+1} = [F_j^{\mathrm{raw}}] - \beta_{\mathrm{out}}[S_{j+1}],$$

and choose the small family E_{j+1} representing this class. Since $S_{j+1}(0) = 0$, equality of normalized totals gives

$$\begin{aligned} F_j^{\mathrm{raw}}(w) &= \beta_{\mathrm{out}} S_{j+1}(w) + E_{j+1}^{\mathrm{raw}}(w) + c_j, \\ c_j &= F_j^{\mathrm{raw}}(0) - E_{j+1}^{\mathrm{raw}}(0). \end{aligned} \tag{63}$$

The equality is on the entire common product domain. Hence the exact updates preserving (62) are

$$\gamma_{j+1} = \gamma_j - \beta_{\mathrm{out}}, \qquad Z_{j+1} = Z_j N_j e^{c_j}. \tag{64}$$

If an input representative is replaced by $E'^{\mathrm{raw}} = E^{\mathrm{raw}} + c$ before integration, replace Z_j by $Z_j e^{-c}$. Both signs are fixed by the raw exponential, not by a convention about the quotient. On real inputs use the compatible real logarithm branches; constant $2\pi i$ changes in another compatible convention do not change the exponentials. No convergence of an extensive vacuum free energy is asserted.

If the couplings are freely prescribed without satisfying (64), the theorem still bounds the corresponding projected maps. It does not then assert repeated factorization of one unchanged density. Along the actual update, the amplitude bounds hold only while the couplings stay in their proved weak range.

For the unprojected value along the envelope,

$$\|\mathcal{F}_j(x_j)\| \le [B_T C_{\mathrm{int}} + C_F + L_{\mathrm{quant}} C_{\mathrm{int}}] b^2, \qquad L_{\mathrm{quant}} = \sup_j L_{\mathrm{quant},j}. \tag{65}$$

Separating $\mathcal{T}_j x_j$, $\mathcal{G}_j(0)$ and $\mathcal{G}_j(x_j) - \mathcal{G}_j(0)$ proves this bound. The finite Maxwell functional consequently gives

$$|\gamma_{j+1} - \gamma_j| \le C_\beta [B_T C_{\mathrm{int}} + C_F + L_{\mathrm{quant}} C_{\mathrm{int}}]. \tag{66}$$

This is uniform in the finite regulator but has no sign. For a specified finite number n of steps, an initial inverse coupling exceeding the weak threshold by n times this bound suffices for weak containment on that horizon. It does not select a desired final coupling, establish an averaged positive drift or control an infinite physical trajectory. Comparing different running-coupling histories would require an additional coupling-parameter comparison.

2.7 Precise closure and remaining obligations

The theorem closes the finite *representative-budget* interface for this homogeneous local scheme. At every allowed stage the chosen connected family has norm $O(b^2)$ at the same carrier exponent and field radius. Every actual next input and Gaussian budget holds; finite normalized-total identities are exact; vacuum constants are compensated by (63)–(64). The q-block amplitude contraction is at most one half for equal prescribed coupling data.

This does not prove smallness of an untouched carrier array, identify a one-shot composite sector with sequential primitive sectors, supply large-field or mixed-history terms, or show that the local germs represent every global Wilson history. Those interpretations require their own orbit-domain and measure identities. Every retained window remains precisely the one in the common-domain theorem.

A physical density trajectory must obey (64) and remain weak. Its ultraviolet tuning and beta drift, other field sectors, vacuum limits, observables, nontriviality and a continuum measure remain separate. None follows solely from the finite representative induction proved here.

3 Coupling parity and the leading zero-input correction

Fix the composite geometry, auxiliary grouping, trace convention and input-independent classical system of the homogeneous local macrostep. Use one common outer field polydisc of radius $\rho > 0$ and connected carrier exponent a_0. All constants below may depend on that fixed geometry and ρ, but are uniform in preceding homogeneous depth and compatible volume. Infinite statements concern the same selected local-limit system. No inherited interaction is present.

The zero-input correction has the exact decomposition

$$\mathcal{G}_{M,g,T}(0;w) = \Delta_M(w) + \mathcal{Q}_{\det,M}(w) + \mathcal{L}_{M,g,T}(w), \qquad \mathcal{L}_{M,g,T} = \log \mathbb{E}_{C_M(w)}[\chi_T(\xi) e^{V_{\text{base}}(w,\xi;g)}]. \quad (67)$$

The pivot term Δ_M, Gaussian determinant term $\mathcal{Q}_{\det,M}$, covariance $C_M(w)$, stationary branch and retained chart are independent of g. The actual cutoff χ_T is a product of centered retained link/color balls. Field-independent powers of g, flat determinants, Haar constants and the compact gauge-insertion normalization belong to the separately tracked vacuum factor. In particular no analyticity of that exact compact normalization at $g = 0$ is assumed.

We use the anchored norm

$$\|E\|_{a_0,\rho} = \sup_i \sum_{H \ni i} e^{a_0|H|} \|E_H\|_\rho$$

and its quotient by equality of vacuum-normalized totals. A coefficient family is not required to be individually gauge invariant. The estimates below refer to this family norm or its indicated quotient, not to an extensive scalar supremum.

3.1 Exact parity of the centered factors

Keep T fixed while allowing complex g. Every nonlinear unwindowed factor is evaluated in the actual retained chart at the same product $g\xi$. For a scalar action or gauge function F, the centered contribution

is

$$-\frac{1}{g^2}\left[F(g\xi) - F(0) - g\,DF(0)\xi - \frac{g^2}{2}D^2F(0)[\xi,\xi]\right]. \tag{68}$$

The apparent singularity at zero is removable: the numerator starts at total fluctuation degree three. Replacing g by $-g$ is exactly the same as replacing ξ by $-\xi$ in this expression. This remains true on the independent centered background extension; it does not rely on dropping a linear term only at a physical stationary background.

The straightening determinant and Haar factors are analytic functions of $g\xi$ with their near-origin logarithms, and obey the same identity. They need not be separately even in ξ. Consequently the exact total base exponent satisfies

$$V_{\text{base}}(w,\xi;-g) = V_{\text{base}}(w,-\xi;g). \tag{69}$$

The centered Gaussian density is even in ξ, also for the allowed complex background covariance. The fixed real window is invariant under $\xi \mapsto -\xi$. This real change of integration variable gives

$$Z_{M,-g,T}(w) = Z_{M,g,T}(w). \tag{70}$$

No contour deformation or change of the cutoff with the background is involved.

3.2 A complex disk uniform in the fixed threshold

The actual local chart has a common analytic neighborhood in $g\xi$. The retained window bounds that variable by a fixed constant times $|g|T$. The centered action, gauge and measure estimates on the outer field domain give

$$\|V_{\text{base}}\|_{\text{strong}} \le C_{\text{base}}(|g|T^3 + |g|T + |g|^2T^2). \tag{71}$$

The strong norm is the one used by the strict Gaussian support and overlap theorem; its constants are fixed before g, T. Choose $T_0 \ge 1$ so the bad-block tail meets that theorem's strict budget, and then choose a sufficiently small fixed $c > 0$. For all $T \ge T_0$, the disk

$$|g| < r_T := cT^{-3} \tag{72}$$

lies inside the local chart and the strict base budgets. Indeed $|g|T \le cT^{-2} \le c$ and the right side of (71) is at most $C_{\text{base}}(2c + c^2)$. The precision and covariance budgets do not move with g.

The strict Gaussian theorem therefore supplies a nonzero integral and its holomorphic logarithm throughout (72), with a connected coefficient family satisfying

$$\sup_{|g|<r_T} \|\mathcal{L}_{M,g,T}\|_{a_0,\rho} \le H_{\mathcal{L}}. \tag{73}$$

Here $H_{\mathcal{L}}$ is independent of M, volume and $T \ge T_0$. Analyticity in g uses the removable remainders in (68), analytic local measure factors, finite Gaussian moments and the normally convergent positive support/cluster majorants. The sharp indicator is fixed and never differentiated.

The logarithm branch is fixed at $g = 0$ by the same connected Gaussian expansion. On the connected disk, the logarithms in (70) differ by a constant multiple of $2\pi i$, and that multiple is zero at $g = 0$. Thus the total logarithm is even.

3.3 An even representative and the quantitative remainder

Theorem 3.1 (Fixed-threshold parity estimate)**.** *Under the preceding actual-chart and strict-budget hypotheses, for $T \geq T_0$ and $|g|T^3 \leq c/2$,*

$$\|[\mathcal{L}_{M,g,T} - \mathcal{L}_{M,0,T}]\|_{\mathcal{Q},a_0,\rho} \leq 2H_{\mathcal{L}} \left(\frac{|g|T^3}{c}\right)^2. \tag{74}$$

Moreover, with

$$D_M := \Delta_M + \mathcal{Q}_{\mathrm{det},M}, \tag{75}$$

there is a uniform constant C_{par} such that

$$\|[\mathcal{G}_{M,g,T}(0) - D_M]\|_{\mathcal{Q},a_0,\rho} \leq C_{\mathrm{par}}[\tau(T) + |g|^2 T^6]. \tag{76}$$

Proof. Evenness of a total does not require the original localized coefficients to be even. Replace each coefficient by

$$\mathcal{L}_H^{\mathrm{ev}}(g) := \frac{\mathcal{L}_H(g) + \mathcal{L}_H(-g)}{2}. \tag{77}$$

This retains its carrier and does not increase (73). Its total is the same logarithm. Subtracting the coefficient family at $g = 0$ represents the actual difference in (74); that generally background-dependent cutoff logarithm is not being declared a vacuum constant.

Banach-valued Cauchy on radii increasing to r_T gives

$$\|\mathcal{L}^{\mathrm{ev}}(g) - \mathcal{L}^{\mathrm{ev}}(0)\|_{a_0,\rho} \leq H_{\mathcal{L}} \sum_{n\geq 1} (|g|/r_T)^{2n} \leq 2H_{\mathcal{L}}(|g|/r_T)^2. \tag{78}$$

The same estimate follows by taking the Cauchy integral before the positive anchored sums. It uses no interchange of an order sum with a root supremum and no count of background coordinates. Taking the quotient proves (74).

At $g = 0$ all centered action and gauge remainders and the $g\xi$ measure remainders vanish. The remaining logarithm is precisely the Gaussian cutoff logarithm, whose strict connected norm is at most $C\tau(T)$. Combine this with (74) and (67). □

3.4 The physical threshold and the Maxwell functional

Only after the fixed-T estimate has been proved do we evaluate it at the prescribed real threshold $T = T(g)$. This is an inequality in two parameters, not a claim that the sharp g-dependent window is holomorphic. For a fixed sufficiently large multiple of $\sqrt{\log(1/g)}$ and a fixed lower threshold,

$$\tau(T(g)) \longrightarrow 0, \qquad g^2 T(g)^6 = O\Big(g^2 \log(1/g)^3\Big).$$

The tail can be made a prescribed positive power of g by its fixed coefficient, subject to the stated Gaussian tail bound.

The fixed-order kernel estimate on the outer field domain gives $\|\beta\| \leq C_\beta \rho^{-2}$. Therefore

$$\Big|\beta(\mathcal{G}_{M,g,T(g)}(0)) - \beta(D_M)\Big| \leq C_\beta \rho^{-2} C_{\mathrm{par}}[\tau(T(g)) + g^2 T(g)^6]. \tag{79}$$

This convergence is uniform in M under the established common bounds. It neither computes $\beta(D_M)$ nor proves that those coefficients converge as M grows.

There is also a useful inner-radius form for other bounded linear marginal coordinates. The actual zero-input totals are locally gauge and globally adjoint invariant. Equation (76) along $T(g)$ shows that

D_M is an invariant total as well, since the invariant subspace of the outer quotient is closed. For a semisimple Lie algebra the total first derivative of their difference vanishes. Remove the degree-zero and degree-one background sectors of a representative of that difference and restrict to $b \le \rho/2$. Homogeneous phase projection gives

$$\|[\mathcal{G}_{M,g,T(g)}(0) - D_M]\|_{\mathcal{Q},a_0,b} \le \frac{2C_{\mathrm{par}}}{\rho^2}\, b^2[\tau(T(g)) + g^2 T(g)^6]. \tag{80}$$

Taking representatives arbitrarily close to the outer quotient infimum justifies the same assertion when the outer estimate was stated only as a quotient bound. No coefficientwise gauge identity is needed.

3.5 Scope

The terms Δ_M and $\mathcal{Q}_{\mathrm{det},M}$ have no small factor g. Their removal of total constant and linear sectors gives a quadratic inner-field-radius bound, not an $O(g^2)$ bound for the full correction. The theorem identifies a concrete leading determinant/pivot functional and a vanishing error. It proves no sign for its Maxwell coefficient, no running-coupling trajectory and no identification of the local sector with the full Wilson measure. All vacuum factors remain separately tracked. The fixed classical-system and common-domain qualifications are unchanged.

4 Exact classical preservation of the Maxwell coefficient

This section concerns the invariant infinite interaction quotient on one selected joint classical background system. The coefficient β_M is the true infinite total-kernel Maxwell moment. It is not the finite symmetric-cutoff ratio $\beta_{M,N}$. Fix the primitive ratio, grouping J, activity exponent a_0, and one positive field radius b. The actual homogeneous reference and projection satisfy

$$\beta_M(S_M) = 1, \qquad \|S_M\|_b \le C_S b^2, \qquad P_M = I - S_M\beta_M, \qquad \|P_M\| \le C_{\mathrm{ext}}(J). \tag{81}$$

The actual pre-gauge substitution has the scalar identities

$$\mathcal{T}_{M,L}S_M = S_{ML}, \qquad \mathcal{T}_{MR,L}\mathcal{T}_{M,R} = \mathcal{T}_{M,RL}. \tag{82}$$

They are identities of normalized interactions, not of chosen support coefficient arrays.

The complete-field pre-gauge family obeys, for every fixed prescribed finite classical reserve A,

$$\sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+A|X|}\|a_{M,L;i,X}\|_{C^{1,\alpha};b} \le C_* b/L, \qquad 0 < b \le r_A, \tag{83}$$

on one outer radius $r_A > 0$ independent of M, L. The entire output fiber and its physical first-gradient Hölder quotients are inside the support sum, and the coefficients agree on overlapping collars. Fix J and the reserves so that $2(a_0+1)/J \le \kappa$ and $A \ge 4(a_0+1)$.

The argument uses countably many arbitrarily large endpoint powers. They can be included in the chosen joint system by a further diagonal extraction with the same r_A; previously fixed limits remain unchanged. The exact finite comparison and controlled gauge identities then give (82) on this enlarged joint system. No independence from unrelated subsequences is asserted.

4.1 The quadratic estimate needs no large field-scaling disk

Let $y = P_M x$ for an invariant class x. Global semisimple color invariance gives $D\Phi_y(0) = 0$. The exact infinite Ward classification and hypercubic symmetry give

$$\Gamma_2^y(p) = O(|p|^4), \qquad \|K_2(y)\|_{\sigma(J),\mathrm{row}} \le C(J)b^{-2}\|y\|_b. \tag{84}$$

The fixed midpoint phases preserve the order of vanishing in tail coordinates. Define $Q_y(A) = K_2(y)[A,A]/2$. For compact coarse w, the pre-gauge image is exponentially decaying and summable, so $Q_y(a_{M,L}(w))$ is well defined.

The quadratic part of the spatial localization proof yields a connected family $Q_{y,L}$ representing that entire substituted quadratic polynomial, with

$$\|Q_{y,L}\|_{a_0,b} \le C_2(J)L^{-\alpha}\|y\|_b, \qquad L \ge L_0(J), \quad 0 < b \le r_A. \tag{85}$$

Indeed the divided-difference factorization is $K_2 = d^-\widetilde{H}d^+$, with $\sum_r \widetilde{H}(r) = 0$, and gives the exact expression

$$\frac{1}{2}\sum_{x,r,\rho,\tau}[D_\rho a(x)]^T\widetilde{H}_{\rho\tau}(r)[D_\tau a(x+r) - D_\tau a(x)].$$

The complete family bounds supply CbL^{-2} for the first difference and $CbL^{-2-\alpha}|r|^\alpha$ for the nearby gradient difference. The exponential kernel tail controls the far terms. The $(JL)^4$ anchor count leaves $L^{-\alpha}$. All constants are fixed after J and before L.

Only $a_{M,L}(w)$ on the original b-ball is used. There is no substitution $a_{M,L}(\lambda w)$ with $|\lambda|$ proportional to L. Thus the high-jet condition $bL \le$ constant is absent from (85).

Although $Q_{y,L}$ need not be invariant as a whole nonlinear functional, the chosen Hessian moment extends to every analytic interaction quotient as a bounded linear functional:

$$|b_{\mathrm{mom}}(Z)| \le C_\beta(J)b^{-2}\|Z\|_b. \tag{86}$$

On invariant classes it is β. This norm bound is just a fixed second derivative and a fixed second spatial moment; it does not require a Ward identity for Z.

4.2 The transported Hessian and the semigroup limit

Since $a_{M,L}(0) = 0$ and $D\Phi_y(0) = 0$, the chain rule gives

$$D^2(\Phi_y \circ a_{M,L})(0)[u,v] = D^2\Phi_y(0)[Da_{M,L}(0)u, Da_{M,L}(0)v]. \tag{87}$$

The right side is also the Hessian of $Q_y \circ a_{M,L}$. Higher input field degrees cannot contribute to degree two, and the possible $D\Phi_y(0)D^2a_{M,L}(0)$ term is zero. Differentiation is justified first on finitely supported directions by local normal convergence, then for the moment by the anchored exponential kernel bounds. Hence

$$\beta_{ML}(\mathcal{T}_{M,L}y) = b_{\mathrm{mom}}(Q_{y,L}). \tag{88}$$

Combining (85) and (86), followed by the exact reference transport, proves

$$|\beta_{ML}(\mathcal{T}_{M,L}x) - \beta_M(x)| \le C_m(J)b^{-2}L^{-\alpha}\|x\|_b. \tag{89}$$

This estimate is uniform in preceding depth, at fixed $b > 0$.

Theorem 4.1 (Exact infinite Maxwell preservation)**.** *For every fixed admissible ratio R and every invariant class x on this common domain,*

$$\beta_{MR}(\mathcal{T}_{M,R}x) = \beta_M(x). \tag{90}$$

Proof. Put $z = \mathcal{T}_{M,R}x$. The fixed-R classical transport is bounded, say $\|z\|_b \le B_R(J)\|x\|_b$. Apply (89) to z at depth MR and endpoint L, and to x at depth M and endpoint RL. By (82), their leading transported moments agree. Therefore

$$|\beta_{MR}(\mathcal{T}_{M,R}x) - \beta_M(x)| \le C_m(J)b^{-2}[B_R(J)L^{-\alpha} + (RL)^{-\alpha}]\|x\|_b. \tag{91}$$

The left side is independent of L. Let L tend through the admissible endpoint powers to infinity in the joint system. □

The proof needs no convergence of the full interactions $\mathcal{T}_{M,L}x$ as $L \to \infty$, and no bound on their whole norms uniform in L. Only the fixed-R map, the scalar semigroup, and the quadratic-only bound are used.

There is no hidden domain shrink. The actual pre-gauge holomorphy and (83) have a fixed primitive-scale radius. A real minimum comparison at one large endpoint can initially require a smaller real germ because physical norms rescale. Both scalar sides of (82), however, are holomorphic on their specified common b-domain; equality on that nonempty germ extends throughout the connected domain. On compact field subspaces the anchored and summable-field bounds justify that continuation. The high-jet dilation radius plays no role. Countably many endpoints can therefore be included without replacing b by a sequence tending to zero.

4.3 The original Maxwell coordinate isolates the leading drift

Equations (90) and (82) imply

$$\beta_{MR}\mathcal{T}_{M,R} = \beta_M, \qquad \mathcal{T}_{M,R}P_M = P_{MR}\mathcal{T}_{M,R}. \tag{92}$$

Thus the classical leakage on $\ker \beta_M$ is exactly zero for this infinite system. An adapted-coordinate correction introduced solely to remove that leakage can be chosen zero; the general coordinate theorem remains valid for other operator families.

For the fixed-coupling quantum map $\mathcal{F}_{M,g,T} = \mathcal{T}_{M,R} + \mathcal{G}_{M,g,T}$, let $x = P_M x$, $\|x\|_b \le C_{\mathrm{int}} b^2$. The parity theorem defines $D_M = \Delta_M + \mathcal{Q}_{\mathrm{det},M}$, as in (75), and gives the inner bound (80). The input Lipschitz estimate gives

$$\|\mathcal{G}_{M,g,T}(x) - \mathcal{G}_{M,g,T}(0)\|_b \le C_{\mathrm{quant}}(gT/b) C_{\mathrm{int}} b^2.$$

Applying the original Maxwell moment, with $\|\beta_{MR}\| \le C_\beta(J) b^{-2}$, yields

$$\begin{aligned} &|\beta_{MR}(\mathcal{F}_{M,g,T}(x)) - \beta_{MR}(D_M)| \\ &\quad \le \frac{2C_\beta C_{\mathrm{par}}}{\rho^2}[\tau(T) + g^2 T^6] + C_\beta C_{\mathrm{quant}} C_{\mathrm{int}}\, gT/b. \end{aligned} \tag{93}$$

All constants are uniform in preceding depth on the fixed classical system. The field radius b is fixed before the real threshold $T = T(g)$ and $g \to 0$ are considered. For an actual density update $\gamma_{\mathrm{next}} = \gamma - \beta_{MR}(\mathcal{F}(x))$, the leading decrement in the original coupling coordinate is therefore $\beta_{MR}(D_M)$, with the displayed vanishing error. No adapted marginal coordinate is needed for this conclusion.

4.4 Finite and physical limitations

The finite coefficient $\beta_{M,N}$ is a short-lift ratio and is not the coefficient in (90). Its Ward defects and folded tails are nonzero at finite period. This argument does not prove exact finite marginal preservation or set the finite classical leakage to zero. The finite reference telescope and finite contraction remain valid without that identity.

Equation (93) does not compute the determinant/pivot coefficient, prove its sign or convergence with M, or control an infinite physical coupling trajectory. The result concerns the selected joint infinite classical system and does not establish subsequence independence, global sector matching, observable convergence or continuum existence.

5 The coupling calculation and the remaining existence argument

The finite local induction supplies actual admissible interaction families at every stage. Its exact density bookkeeping is

$$\gamma_{j+1} = \gamma_j - \beta_{j+1}(\mathcal{F}_{j,g_j}(E_j)), \qquad \gamma_j = g_j^{-2}.$$

A trajectory of one physical density must obey this equation. The bounded magnitude of its increments does not determine their sign or tune an arbitrarily long ultraviolet sequence to a fixed interacting endpoint.

The parity theorem isolates a concrete leading functional,

$$D_M = \Delta_M + \mathcal{Q}_{\det,M}.$$

Exact infinite classical transport and the quantum input bound reduce the drift on the $O(b^2)$ amplitude envelope to

$$\beta_{j+1}(D_{M_j}) + O\left(\tau(T(g_j)) + g_j^2 T(g_j)^6 + \frac{g_j T(g_j)}{b}\right).$$

Here $b > 0$ is fixed before sending the coupling to zero; constants may depend on that fixed field radius and geometry. This expression concerns the infinite quotient, with the finite cutoff-extraction distinction retained above. The leading Maxwell coefficient must be evaluated using the actual stationary background, pivot coordinates, gauge normalization and retained Gaussian. Every implicit background derivative belongs to that calculation. A universal perturbative coefficient from another scheme cannot substitute for this identification.

The next measure-theoretic steps remain substantial. The homogeneous local germ must be represented on the required gauge-orbit domains, and mixed and large-field histories must be incorporated with their exact measure identities. Uniform source estimates must then produce a nontrivial continuum observable hierarchy with the symmetry, positivity and reconstruction properties specified in Notebook 01. No conclusion about those steps follows merely from changing interaction representatives or bounding a finite local integral.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–30 in this workspace. The finite ordinary minimum, common-domain local quantum map, total Ward extraction and quantum quotient estimates are the principal inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Commun. Math. Phys. **109** (1987), 249–301. Publisher record.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22. Publisher record. The finite quotient and representative arguments here are given separately; the published induction theorem is not being imported with altered hypotheses.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.